

Retail Growth & Inventory Optimization Strategy for Maven Toys

Executive Summary

Maven Toys is a multi-city toy retail chain operating across Mexico with a broad portfolio of products spanning Toys, Games, Electronics, Sports & Outdoors, and Art & Crafts categories.

The company sought to improve retail performance and inventory efficiency by analyzing historical sales, product performance, store operations, and inventory allocation patterns. Despite consistent revenue generation, management identified operational concerns including uneven inventory distribution, stockout risk for high-demand products, excess inventory for low-performing items, and inconsistent store-level performance across cities.

This project combines Retail Analytics, Inventory Analytics, and Supply Chain Analytics to provide actionable business insights and strategic recommendations.

Using PostgreSQL for advanced SQL analysis and Power BI for dashboarding and business intelligence visualization, the analysis identified high-performing products, profitable categories, regional demand variations, inventory inefficiencies, and supply chain optimization opportunities.

Key findings suggest that a small percentage of products contribute disproportionately to total revenue, while several products experience high demand relative to available inventory levels. The analysis also identified substantial differences in revenue generation across store locations and cities, enabling targeted operational and inventory strategies.

The project ultimately provides a data-driven framework for improving inventory turnover, reducing stockout risk, optimizing product allocation, and increasing operational profitability.

PART 1:

1. Business Problem & Objective

Maven Toys operates multiple retail stores across several Mexican cities and manages inventory for dozens of product lines. As the business expanded, management faced increasing challenges related to inventory allocation, demand variability, and store-level performance optimization.

The company identified several key operational concerns:

- Some high-demand products frequently experienced inventory shortages.
- Several stores carried excess inventory with relatively low sales velocity.
- Product demand patterns varied significantly by region and store location type.
- Management lacked centralized visibility into revenue, profitability, and inventory efficiency.
- Inventory replenishment decisions were not fully aligned with historical demand trends.
- Certain products generated strong sales but relatively low profitability.

Without a data-driven decision-making framework, the company risked:

- Lost sales opportunities
- Increased holding costs
- Reduced inventory turnover
- Poor inventory allocation
- Lower operational efficiency

The objective of this project was to analyze retail sales and inventory data to identify operational inefficiencies and provide strategic recommendations for business growth and inventory optimization.

The primary objectives of this project were:

- Analyze sales performance across products, stores, and cities.
 - Identify high-performing and low-performing products.
 - Measure product profitability and category contribution.
 - Evaluate store-level operational performance.
 - Analyze inventory allocation and stock availability.
 - Identify products with elevated stockout risk.
 - Detect overstocked inventory segments.
 - Develop actionable supply chain and inventory optimization recommendations.
 - Build interactive executive dashboards for decision support.
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2. Dataset Overview

The project utilized the Maven Toys retail dataset containing transactional sales, product, store, and inventory information.

Tables Used

Sales Table

Contains approximately 800,000 transaction records including:

- Sale ID, Date, Store ID, Product ID, Units Sold

Products Table

Contains product information including:

- Product Name, Product Category, Product Cost, Product Price

Stores Table

Contains store metadata including:

- Store Name, Store City, Store Location Type, Store Open Date

Inventory Table

Contains current inventory snapshots including: Store ID, Product ID, Stock on Hand

3. Tools & Technologies Used

PostgreSQL

Used for:

- Data cleaning
- Data validation
- Exploratory analysis
- Advanced SQL analytics
- Business metric generation
- Joins
- Aggregations
- Common Table Expressions (CTEs)
- Window Functions
- Ranking Functions
- Running Totals
- Growth Analysis
- Inventory Risk Calculations

Power BI

Used for:

- Data modeling
 - DAX calculations
 - Interactive dashboards
 - KPI visualization
 - Executive reporting
 - Revenue
 - Profit
 - Profit Margin
 - Inventory Turnover
 - Stock Coverage
 - Demand-to-Stock Ratio
 - Revenue Contribution %
 - ABC Classification
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PART 2:

5. KPIs

Total Revenue: \$14.44 M

Total Profit: \$ 4.01 M

Total Units Sold: 1.09 M

Profit Margin: 27.79%

Inventory Units: 30K

Total Products: 35

Total Cities: 29

Total Stores 50

Top Selling Product: Lego Bricks

Average Selling Price: \$13.25

Total Inventory Value: 300K

Stock Coverage Days: 17

Inventory Turnover: 36.67

6. Revenue & Sales Analysis

The sales analysis identified key revenue drivers across products, stores, and cities.

Key Findings

- Total company revenue exceeded \$14 million during the analysis period.

- Revenue remained usually steady with a few sudden dips and spikes.
- A limited number (nearly 40%) of products contributed a disproportionately high share (80%) of overall revenue.
- Downtown stores consistently outperformed Residential stores in both revenue and profitability.
- Just 7 cities (25%) generate 50% of the total revenue.

Product Category Performance

- Toys category alone contributed approximately 35% of total revenue.
- Games and Electronics categories demonstrated strong profit margins 30% and 44% respectively.
- While the sales of Electronics have continuously plummeted, Art & Crafts are the only products that have shown a consistent growing demand.

Seasonal Trends

- Revenue increased significantly during Q4 holiday period.
 - Several product categories exhibited strong seasonal demand spikes. E.g. Toys and Sports-outdoors in March to June.
 - Monthly demand patterns suggested opportunities for seasonal inventory planning.
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7. Store Performance Analysis

Store-level analysis revealed major operational differences across locations and store types.

Key Findings

- Although the difference between the most-earning store and the least-earning store is 4 times, there is not a significant variance in the spread of revenue generated by the stores.
- Airport stores achieved the highest average revenue per store.
- Airport stores demonstrated stronger inventory turnover compared to other locations.
- There was not enough evidence to suggest a relation between the time since the opening of the store and its revenue.

Operational Insights

- Revenue was distributed largely uniformly over most of the stores.
- Several underperforming stores maintained relatively high inventory levels despite lower sales activity.
- Store location type seems to have influenced customer purchasing behaviour.

8. Product & Profitability Analysis

The product analysis identified the products and categories driving both revenue and profitability.

Key Findings

- A small percentage of products (40%) contributed the majority(80%) of the total revenue.
- Certain products like Lego Bricks, Magic Sand and Rubik's Cube, generated high sales volume but relatively low margins.
- High-margin products like Mini Basketball hoop, Plush Poney, Etch a Sketch, represented opportunities for targeted promotional strategies.
- Some low-revenue products consumed excessive inventory space.

Profitability Insights

- Electronics category achieved one of the strongest average profit margins of about 45%.
- Electronics products demonstrated consistent loss in revenue and sales..
- Several products showed strong profitability despite moderate sales volumes.

Product Segmentation

Products were segmented into:

- High Revenue / High Profit
- High Revenue / Low Profit
- Low Revenue / High Profit
- Low Revenue / Low Profit

This segmentation enabled strategic product prioritization.

9. Inventory & Supply Chain Analysis

The inventory analysis focused on stock availability, turnover efficiency, and inventory allocation optimization.

Key Findings

- Several high-demand products like Action Figure maintained critically low inventory levels.
 - Certain low-demand products like Dinosaur Figures remained significantly overstocked.
 - Inventory allocation was uneven across stores and regions.
 - Inventory turnover varied substantially across product categories.
 - Some stores carried excessive inventory relative to actual sales demand.
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10. ABC Inventory Classification

ABC analysis was performed to classify products based on revenue contribution.

Classification Logic

Class A

Top revenue-generating products contributing approximately 70% of total revenue.

Class B

Moderate contribution products contributing approximately 20% of revenue.

Class C

Low-contribution products representing the remaining revenue share.

Key Findings

- Class A products represented a relatively small percentage of the product portfolio.
 - Class A products required tighter inventory control and replenishment prioritization.
 - Several Class C products maintained disproportionately high inventory levels.
 - Inventory optimization opportunities existed primarily within lower-performing product groups.
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11. Pareto Analysis

Pareto analysis demonstrated that approximately 40% of products generated nearly 80% of total revenue.

Key Insights

- Revenue concentration was highly uneven.
 - A limited set of products drove the majority of business performance.
 - Strategic focus on high-performing products could significantly improve profitability and inventory efficiency.
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12. Key Business Insights

Revenue Insights

- Revenue growth remained concentrated in a small number of product categories.
- Downtown stores consistently generated superior performance metrics.

Inventory Insights

- Several high-demand products faced elevated stockout risk.
- Overstock conditions existed within low-performing product categories.
- Inventory allocation did not consistently align with historical demand patterns.

Operational Insights

- Inventory turnover efficiency differed significantly across product categories.
 - Product profitability and sales volume were not always aligned.
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13. Strategic Recommendations

Recommendation 1 — Increase Inventory for High-Demand Products

Increase inventory allocation for products demonstrating:

- High demand
- Strong turnover

- Elevated stockout risk

Expected Impact:

- Reduced lost sales
 - Improved customer satisfaction
 - Increased revenue capture
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Recommendation 2 — Reduce Overstocked Inventory

Reduce inventory exposure for:

- Low-turnover products
- Low-contribution Class C products
- Underperforming categories

Expected Impact:

- Lower holding costs
 - Improved inventory efficiency
 - Better working capital utilization
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Recommendation 3 — Implement Regional Inventory Strategies

Adopt city-specific inventory allocation strategies based on localized demand patterns.

Expected Impact:

- Improved inventory accuracy
 - Reduced regional stock imbalances
 - Enhanced operational responsiveness
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Recommendation 4 — Prioritize Class A Products

Prioritize:

- replenishment
 - marketing
 - shelf allocation
- for Class A products.

Expected Impact:

- Higher revenue growth
 - Better inventory utilization
 - Increased operational focus
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Recommendation 5 — Improve Demand Forecasting

Utilize historical sales trends and seasonality patterns for future inventory planning.

Expected Impact:

- Better procurement decisions
 - Reduced stockout frequency
 - Improved inventory planning accuracy
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Recommendation 6 — Optimize Store-Level Inventory

Rebalance inventory allocation based on:

- store demand
- sales velocity
- regional performance

Expected Impact:

- Lower excess inventory
 - Improved stock availability
 - Enhanced operational efficiency
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14. Conclusion

This project successfully combined Retail Analytics, Inventory Analytics, and Supply Chain Analytics to deliver actionable business intelligence for Maven Toys.

The analysis identified major opportunities for:

- revenue optimization
- inventory efficiency improvement
- operational enhancement
- strategic inventory allocation

The findings demonstrated that a data-driven approach can significantly improve inventory utilization, reduce operational risk, and support long-term retail growth.

By integrating SQL analytics, Power BI visualization, and business strategy recommendations, the project provides a scalable framework for retail performance optimization and operational decision-making.
